

AWS Academy Cloud Foundations - Exam Preparation

February 23, 2026

Question 1

Explain the difference between AWS Regions and Availability Zones. Use a simple diagram to illustrate how multiple AZs within a Region provide high availability.

Answer

AWS Region is a large geographical area that contains multiple Availability Zones.

An Availability Zone is one or more isolated data centers within a Region.

Multiple AZs in a Region are connected by high-speed private networking and provide high availability because if one AZ fails, applications continue running in other AZs.

How Multiple AZs in a Region Provide High Availability

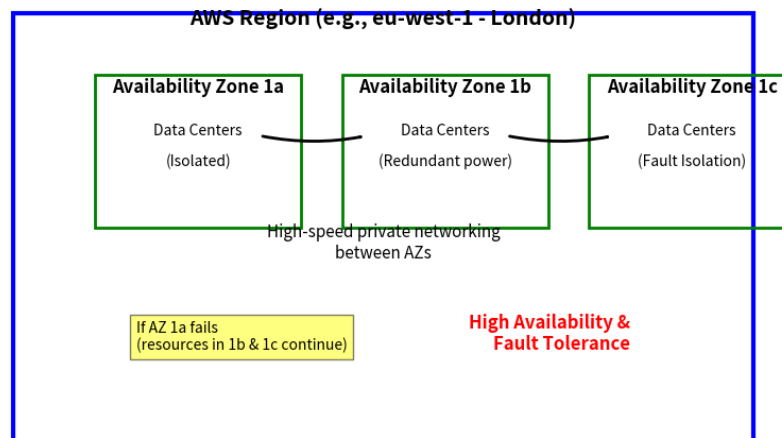


Figure 1: Simple diagram showing how multiple Availability Zones (AZs) inside one AWS Region provide high availability

Question 2

In the context of the AWS Shared Responsibility Model, identify which of the following responsibilities belong to AWS and which to the customer. Justify your answer:

- Applying security patches to an EC2 instance
- Physical security of AWS data centers
- Configuring IAM policies for users
- Oracle upgrades if the Oracle instance runs as an Amazon RDS instance
- Securing the SSH keys to remote access through CLI or SDK

Answer

The AWS Shared Responsibility Model divides security responsibilities between AWS and the customer. AWS is responsible for **security of the cloud**, which includes the physical infrastructure, hardware, and managed services. The customer is responsible for **security in the cloud**, which includes operating systems, configurations, credentials, and access management.

1. Applying security patches to an EC2 instance – Customer Responsibility

Amazon EC2 is an Infrastructure as a Service (IaaS) offering, where the customer has full control over the operating system. Therefore, the customer is responsible for applying security patches, updates, and maintenance to the EC2 instance operating system.

2. Physical security of AWS data centers – AWS Responsibility

AWS is responsible for protecting the physical infrastructure of the cloud, including data centers, servers, networking hardware, and facilities. This includes physical access control, surveillance, and environmental protection.

3. Configuring IAM policies for users – Customer Responsibility

Identity and Access Management (IAM) configuration is the responsibility of the customer. The customer controls who can access AWS resources and what actions they are allowed to perform by defining IAM users, roles, and policies.

4. Oracle upgrades when running as Amazon RDS – AWS Responsibility

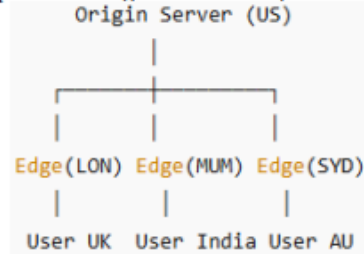
Amazon RDS is a managed service where AWS handles database maintenance tasks such as patching, upgrades, and backups. Therefore, Oracle database upgrades in Amazon RDS are managed by AWS.

5. Securing SSH keys for remote access (CLI/SDK) – Customer Responsibility

SSH keys are authentication credentials used to access EC2 instances and AWS services. The customer is responsible for securely storing, managing, and protecting these keys from unauthorized access.

In conclusion, AWS is responsible for protecting the infrastructure that runs cloud services, while the customer is responsible for securing their data, configurations, operating systems, and access controls within the cloud.

Consider the diagram below showing an **AWS CloudFront distribution**. What is **CloudFront** and **Edge Location**? Explain how **Edge Locations** help reduce latency for global users.



Question 3

Answer

Amazon CloudFront is a Content Delivery Network (CDN) service provided by AWS that delivers web content such as web pages, images, videos, and APIs to users with low latency and high transfer speed. CloudFront distributes content through a worldwide network of data centers called Edge Locations.

Edge Location is a physical data center in a specific geographic location where CloudFront caches copies of content closer to end users. These locations are part of AWS's global network and are designed to deliver content quickly to users by reducing the distance between the user and the data source.

How Edge Locations Reduce Latency

When a user requests content, the request is routed to the nearest Edge Location rather than the origin server. If the requested content is already cached at that Edge Location, it is delivered immediately to the user. This reduces the time taken to fetch data from the distant origin server, thereby lowering latency and improving performance.

If the content is not available in the cache, the Edge Location retrieves it from the origin server, stores a cached copy, and then delivers it to the user. Future requests from nearby users are served directly from the Edge Location, resulting in faster response times and reduced load on the origin server.

Example Based on the Diagram

In the diagram, the origin server is located in the US, and Edge Locations are present in London (LON), Mumbai (MUM), and Sydney (SYD). Users in the UK, India, and Australia access content from the nearest Edge Location instead of directly from the US origin server. This minimizes network delay and ensures faster content delivery.

Key Benefits

- Reduced latency due to shorter distance between user and cached content
- Faster content delivery for global users
- Reduced traffic and load on the origin server
- Improved availability and performance

In conclusion, Amazon CloudFront improves global content delivery by caching data at Edge Locations close to users, which reduces latency, increases speed, and enhances overall user experience.

Question 4

A company wants to grant temporary access for its EC2 instance to read/write objects in an S3 bucket without embedding permanent credentials.

- Which IAM component should be used (User, Group, or Role)?
- Briefly explain why it is the most secure choice.

Answer

IAM Component to Use: IAM Role

An **IAM Role** should be used to grant temporary access to the EC2 instance.

Explanation

An IAM Role is designed to provide temporary security credentials to AWS services such as Amazon EC2. When an IAM Role is attached to an EC2 instance, AWS automatically provides temporary credentials through the instance metadata service.

This is the most secure choice for the following reasons:

- **No embedded credentials:** There is no need to store permanent access keys inside the application or EC2 instance.
- **Temporary credentials:** AWS automatically generates and rotates short-term credentials.
- **Reduced security risk:** If credentials are compromised, they expire automatically.
- **Best practice:** AWS recommends using IAM Roles for EC2-to-S3 access instead of IAM users.

Why Not IAM User or Group?

- An **IAM User** requires permanent access keys, which must be stored in the EC2 instance and increases security risk.
- An **IAM Group** is only used to assign permissions to IAM users and cannot be attached directly to EC2 instances.

In conclusion, an IAM Role is the most secure option because it provides temporary, automatically rotated credentials without exposing long-term access keys, ensuring secure and controlled access to the S3 bucket.

Question 5

AWS organizes its services into categories such as Compute, Storage, and Database. Choose one service from each category and explain how these three can work together to support your university's Academic Information Management Application (PUAIS). Draw a block diagram showing the interaction among Compute, Storage, and Database services.

Answer

To support the Academic Information Management Application (PUAIS), we can combine AWS Compute, Storage, and Database services to build a scalable, secure, and reliable system.

Selected AWS Services

- **Compute Service: Amazon EC2** – Used to host the application server and handle user requests such as login, course registration, and result viewing.
- **Storage Service: Amazon S3** – Used to store static files such as student documents, assignments, reports, and uploaded files.
- **Database Service: Amazon RDS** – Used to store structured data such as student records, course information, grades, and faculty details.

How They Work Together

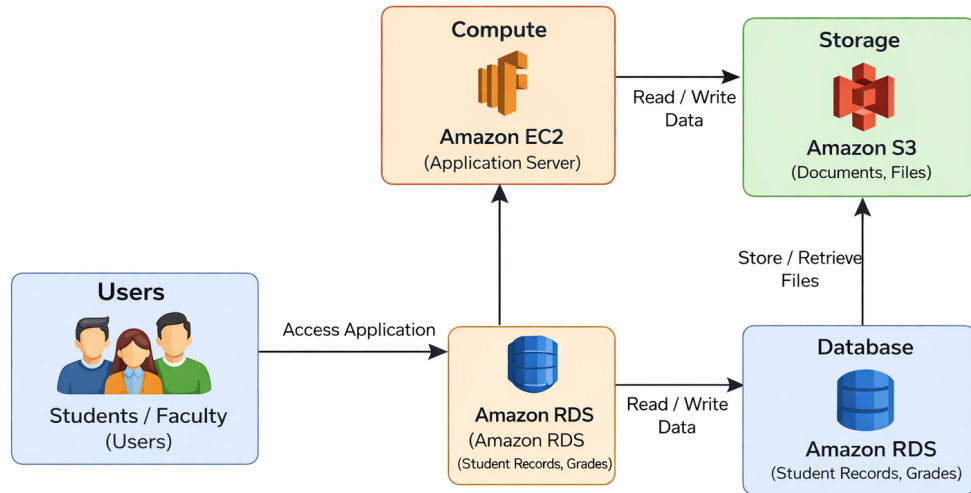
When a user accesses the PAUIS system through a web interface, the request is processed by the application running on the Amazon EC2 instance. The EC2 instance retrieves and updates structured academic data (such as student profiles, course registrations, and grades) from the Amazon RDS database. For file storage, such as student documents or uploaded assignments, the EC2 instance interacts with Amazon S3 to store and retrieve files. This integration ensures efficient processing, reliable storage, and secure data management.

System Workflow

- Users connect to the application hosted on EC2.
- EC2 communicates with Amazon RDS to read/write academic data.
- EC2 stores and retrieves documents from Amazon S3.
- The combined system provides scalable and highly available academic services.

In conclusion, Amazon EC2 handles application processing, Amazon RDS manages structured academic data, and Amazon S3 stores documents and files. Together, these services provide a scalable, secure, and efficient architecture for the Academic Information Management Application (PUAIS).

Interaction between Compute, Storage, and Database services
supporting the Academic Information Management Application (PUAIS)



Interaction between Compute (EC2), Database (RDS), and Storage (S3)
supporting the Academic Information Management Application

AWS Service Categories – Quick Summary Table for Revision

Category	Core Examples	What It Does (Simple)
Compute	EC2, Lambda, ECS/EKS	Run applications and code
Storage	S3, EBS, EFS	Store files, disks, and shared storage
Databases	RDS, DynamoDB, Aurora	Managed databases (SQL and NoSQL)
Networking & Content Delivery	VPC, Route 53, CloudFront	Connect services and deliver content fast
Security & Identity	IAM, KMS, GuardDuty	Control access and encrypt data
Management & Governance	CloudWatch, CloudFormation	Monitor and manage infrastructure
Analytics	Athena, Redshift, Kinesis	Analyze large-scale data
Machine Learning	SageMaker, Rekognition	Add AI and machine learning features

Table 1: Simple revision table of AWS service categories

Question

Setting up a new AWS account requires several security best practices. Describe the steps you would follow to secure a newly created AWS account. Your answer should include:

- Proper use of the root account
- Creation and management of IAM users and groups
- Use of Multi-Factor Authentication (MFA)
- Monitoring and logging with CloudTrail

Why is it considered dangerous to use the root account for day-to-day operations? Provide justification.

Answer

Securing a newly created AWS account is very important to protect cloud resources, data, and services from unauthorized access. The following best practices should be followed:

1. Proper Use of the Root Account

The root account has full administrative access to all AWS services and resources. It should only be used for essential tasks such as initial account setup, billing configuration, and enabling security features. After setup, the root account credentials should be stored securely and not used for regular operations.

2. Creation and Management of IAM Users and Groups

Instead of using the root account, individual IAM users should be created for administrators and other users. IAM groups should be used to assign permissions based on roles (for example, Admin, Developer, or Read-only). Permissions should follow the principle of least privilege, meaning users receive only the access necessary to perform their tasks.

3. Use of Multi-Factor Authentication (MFA)

Multi-Factor Authentication should be enabled for the root account and all IAM users, especially administrators. MFA adds an extra layer of security by requiring a second authentication factor (such as a one-time code from a mobile device) in addition to the password.

4. Monitoring and Logging with AWS CloudTrail

AWS CloudTrail should be enabled to record all API activity and user actions in the AWS account. CloudTrail logs help monitor account activity, detect suspicious behavior, and support security auditing and compliance requirements.

Why Using the Root Account for Daily Operations is Dangerous

Using the root account for everyday tasks is risky because:

- The root account has unrestricted access to all AWS resources.
- If the root credentials are compromised, the entire AWS account can be taken over.
- The root account cannot be restricted by IAM policies.
- Accidental changes or deletions can cause serious damage to cloud resources.

Justification

For security reasons, AWS recommends using the root account only for critical administrative tasks and performing daily operations through IAM users with limited permissions. This approach reduces security risks, prevents unauthorized access, and protects the overall cloud environment.

In conclusion, securing a new AWS account involves protecting the root account, creating IAM users and groups with limited permissions, enabling MFA, and monitoring activity using CloudTrail. These practices ensure strong security and reduce the risk of unauthorized access or data loss.